

## RESEARCH SUMMARY & INTERESTS

---

### *Professional Summary & Highlights*

- **Robotics Hardware Design Expert:** Specialized in developing novel sensors and actuators for service robots, with a strong focus on engineering designs that satisfy diverse operational requirements and multi-functionality. These innovations served as the core hardware foundation for a total of **6 journal publications**, including 4 in *IEEE RA-L*, as well as *IEEE T-RO* and *Nature Communications*.
  - *Pneumatic Artificial Muscles (PAM)*: Developed *ffPAM* designed for compactness and rapid response, and *Double-SpringPAM* designed for bidirectional actuation and self-sensorization.
  - *Vision-Based Tactile Sensors*: Designed *UVtac*, leveraging UV markers for switchable functionalities, and *UVDtact*, achieving both a human fingertip form factor and multi-functional sensing capabilities.
- **Proven Design Capability:** Highly recognized for hardware innovation, winning the **Best Paper Award in Service Robotics at ICRA 2021** (awarded to 12 out of 4,005 papers) for designing a compact flat fabric pneumatic artificial muscle (*ffPAM*).
- **System Integration & Automation:** Extensive hands-on experience in constructing and configuring fully automated experimental setups utilizing industrial robot arms (e.g., UR5e) and motorized XYZ stages.

### *Core Research Areas*

- **Vision-Based Tactile Sensors & Perception:** Hardware design and fabrication of optical tactile sensors, tactile image processing, shape reconstruction, and multi-axis force estimation.
- **Tactile Intelligence & Robotic Automation:** System configuration integrating tactile sensors with robotic arms, and the execution of automated manipulation tasks.
- **Pneumatic Artificial Muscles (PAM) for Robotic Devices:** Pneumatic actuator design, bidirectional actuation mechanism, sensorization, and standalone pneumatic control modules.

## EDUCATION

---

### **Korea Advanced Institute of Science and Technology**

Ph.D. in Mechanical Engineering, GPA: 3.99/4.30, Advisor: Prof. Jung Kim

Daejeon, South Korea

Mar. 2021 –Feb.2026

- Thesis: “A Vision-Based Tactile Sensor with a Fingertip Form Factor for Shape Reconstruction and Force Estimation”

### **Korea Advanced Institute of Science and Technology**

M.S. in Mechanical Engineering, GPA: 3.90/4.30, Advisor: Prof. Jung Kim

Daejeon, South Korea

Mar. 2019 –Feb. 2021

- Thesis: “Design of a Compact Flat Fabric Pneumatic Artificial Muscle for Soft Wearable Robotic Devices”

### **Korea Advanced Institute of Science and Technology**

B.S. in Mechanical Engineering (Minor: Electrical Engineering), GPA: 3.74/4.30

Daejeon, South Korea

Mar. 2014 –Feb. 2019

## PROFESSIONAL EXPERIENCE

---

### Postdoctoral researcher

Korea Advanced Institute of Science and Technology  
InnoCORE PRISM-AI Center & Biorobotics Lab, Advisor: Prof. Jung Kim

Daejeon, South Korea

Mar. 2026 –Current

- Topic: “Tactile Intelligence and Humanoid Tactile Robotic Fingertips for Dexterous Manipulation”

## PUBLICATIONS

---

### JOURNAL ARTICLES

#### [J6] Field-Programmable Robotic Folding Sheet

Hyunkyu Park, Yongrok Jeong, **Woojong Kim**, Junrak Choi, Junseong Ahn, Jun-Ho Jeong, Inkyu Park, and Jung Kim

*Nature Communications* (2025)

(\*\* Nature publisher highlights, Media highlights including CNN)

#### [J5] Graph-Structured Super-Resolution for Geometry-Generalized Tomographic Tactile Sensing: Application to Humanoid Faces

Hyunkyu Park, **Woojong Kim**, Sangha Jeon, Youngjin Na, and Jung Kim

*IEEE Transactions on Robotics* (2024)

(\*\* T-RO editorial highlights)

presented at *ICRA 2025*, Atlanta, USA

#### [J4] Bidirectional Double-Spring Pneumatic Artificial Muscle with Inductive Self-Sensing

Yeonha Cho <sup>†</sup>, **Woojong Kim** <sup>†</sup>, Hyunkyu Park, Jung Kim, and Youngjin Na

(<sup>†</sup> **Equally contributed**)

*IEEE Robotics and Automation Letters*, 8.12 (2023)

#### [J3] Marker-Embedded Tactile Image Generation via Generative Adversarial Networks

Won Dong Kim, Sanghoon Yang, **Woojong Kim**, Jeong-Jung Kim, and Jung Kim

*IEEE Robotics and Automation Letters*, 8.8 (2023)

presented at *ICRA 2024*, Yokohama, Japan

#### [J2] UVtac: Switchable UV Marker-Based Tactile Sensing Finger for Effective Force Estimation and Object Localization

**Woojong Kim** <sup>†</sup>, Won Dong Kim <sup>†</sup>, Jeong-Jung Kim, Chang-Hyun Kim, and Jung Kim

(<sup>†</sup> **Equally contributed**)

*IEEE Robotics and Automation Letters*, 7.3 (2022)

presented at *ICRA 2023*, London, UK

#### [J1] Compact Flat Fabric Pneumatic Artificial Muscle (ffPAM) for Soft Wearable Robotic Devices

**Woojong Kim**, Hyunkyu Park, and Jung Kim

*IEEE Robotics and Automation Letters*, 6.2 (2021)

(\*\* Best Paper Award in service robotics at IEEE ICRA 2021)

presented at *ICRA 2021*, Xi'an, China

### CONFERENCE PAPERS

#### [C3] Robotic Skin with Sparsely Embedded Transducers for Contact Localization on Non-planar Surfaces

Minju Kim, Junhwi Cho, Sangha Jeon, Hyunki Park, **Woojong Kim**, and Jung Kim

2026 23rd International Conference on Ubiquitous Robots (UR), **Accept**

#### [C2] UVDtact: UV Marker-Embedded Fingertip-Like Vision-Based Tactile Sensor for Shape Reconstruction and Force Estimation

**Woojong Kim**, Won Dong Kim, Hyunkyu Park, Joonho Lee, Jeong-Jung Kim, and Jung Kim

2026 IEEE International Conference on Robotics and Automation (ICRA), **Accept**

#### [C1] Vision-Based Aided-Grasping in Teleoperation with Multiple Unknown Objects

Yunjoo Kim, **Woojong Kim**, Seongwoong Hong, Seulki Kyeong, Jirou Feng, and Jung Kim

Companion of the 2020 ACM/IEEE International Conference on Human-Robot Interaction (HRI) (2020)

## IN PREPARATION

### [I3] Tactile-Augmented Vision-Language-Action Framework for Precision Robotic Drawing

Juhee Park <sup>†</sup>, **Woojong Kim** <sup>†</sup>, Gunhee Park, and Jung Kim  
*IEEE Robotics and Automation Letters*

### [I2] Full-Anterior-Mechanoreceptive Anthropomorphic Hand with Super-Resolution Tomographic Tactile Skin

Sangha Jeon, **Woojong Kim**, and Jung Kim  
*Nature Communications*

### [I1] A Proprioceptive Robot Hand Integrated with Vision-Based Tactile Sensors

Hyunjo Chung, **Woojong Kim**, and Jung Kim  
*IEEE Robotics and Automation Letters*

## AWARDS, HONORS, AND FELLOWSHIPS

---

- **PRISM-AI InnoCORE Fellowship** Mar. 2026 - Current  
from Korean Government
- **Invited Speaker at Korea Institute of Industrial Technology (KITECH)** 2026  
*Talk entitled: “Human Fingertip-Like Vision-Based Tactile Sensor for Dexterous Manipulation”*
- **Invited Speaker at the KRoC Flagship Conferences** 2022  
The 17th Korea Robotics Society Annual Conference (KRoC 2022)  
*For the contents of the awarded paper at ICRA 2021*
- **Best Paper Award in Service Robotics (Winner, 12/4005) (ICRA 2021)** 2021  
2021 IEEE International Conference on Robotics and Automation (ICRA 2021)  
*For the paper entitled “Compact Flat Fabric Pneumatic Artificial Muscle (ffPAM) for Soft Wearable Robotic Devices”*
- **KAIST Support Scholarship** Mar. 2019 - Feb. 2025  
from KAIST
- **National Science and Engineering Undergraduate Scholarship** Mar. 2016 - Feb. 2019  
from Korean Government
- **Dean’s List** Fall 2017  
from KAIST College of Engineering
- **Outstanding Achievement Award** 2016, 2015  
from Department of Mechanical Engineering, KAIST

## PROJECTS

---

### Enhancing dexterous robotic tool use via tactile-integrated VLA models

- Project mentor Mar. 2026 –Current
- **Project objective** : R&D on integrating tactile sensing into Vision-Language-Action (VLA) models to enhance precision in robotic manipulation tasks, specifically focusing on drawing tasks using a gripped pen.
  - **Role & Contribution** : Project mentor. Serving as a post-doc mentor to supervise junior researchers and direct the overall research direction of the project. Involved in designing the hardware setup, including a robot arm and a gripper equipped with GelSight sensors, and overseeing the integration of the visuo-tactile framework.

### Development of drone perching mechanism

Person in charge Jan. 2023 –Dec. 2025

- **Project objective** : R&D on the implementation of perching mechanism based on compliant actuators to perform long-term operation of military drones.
- **Funding** : Defense Acquisition Program Administration.
- **Role & Contribution** : Person in charge. Involvement in designing a bistable pneumatic actuator for perching drones, developing a lightweight standalone pneumatic pressure generation & control module, and demonstrating autonomous landing and perching.

## Development of vision-based tactile sensors and tactile image generation

Project participant

Jan. 2021 –Dec. 2023

- **Collaboration** : Korea Institute of Machinery & Materials (KIMM).
- **Project objective** : Collaborative research on the hardware development of vision-based tactile sensors and the data generation of tactile images.
- **Role & Contribution** : Project participant. Involvement in developing a vision-based tactile sensor capable of obtaining independent, switchable tactile images using UV markers, designing tactile signal processing algorithms, and demonstrating robotic tasks including contact force-tracking and object alignment. Contributed to conceptual discussions on tactile image generation strategies. Research paper publication ([J2], [J3]).

## Wearable soft robotic suit for elders

Project participant

Mar. 2019 –Jun. 2021

- **Collaboration** : Korea Institute of Robotics & Technology Convergence (KIRO).
- **Project objective** : National R&D on the soft wearable robotic suit capable of monitoring and assisting the walking of elders.
- **Funding** : Institute for Information & communication Technology Planning & evaluation.
- **Role & Contribution** : Project participant. Involvement in development of wearable sensing suit and participating in discussions on enhancing capacitive strain sensor design.

## Wearable foot drop assistance suit for drop foot patients

Project participant

Mar. 2019 –Dec. 2019

- **Collaboration** : Seoul National University Hospital (Prof. Dong Yeon Lee), Korea Institute of Industrial Technology (KITECH).
- **Project objective** : National R&D on the pneumatic wearable robotic suit for assisting drop foot disorder.
- **Funding** : Ministry of Science and ICT.
- **Role & Contribution** : Project participant. Involvement in developing a pneumatic control module, participating in discussions on thermal management for the battery pack, and performing human subject experiments.

## TECHNICAL SKILLS

---

**Deep learning and computer language:** Pytorch, Python, MATLAB, LaTeX

**3D modeling and simulation:** SolidWorks, ABAQUS

**Hardware prototyping:** Pneumatic actuator design, Soft material manufacturing, 3D printing, Laser cutting, Circuit design

**Robot operation:** LabVIEW, ROS

**Biosignal processing:** Surface electromyography

## DOMESTIC PUBLICATIONS

---

- [8] **Development of Underactuated Finger Mechanism for Vision-Based Tactile Sensor**  
Hyunjo Chung, **Woojong Kim**, Junhwi Cho, and Jung Kim  
*The 20th Korea Robotics Society Annual Conference (KRoC 2025), Korea Robotics Society, 2025*
- [7] **Development of a Fabric-Based Pneumatic Bistable Gripper for Drone Perching**  
**Woojong Kim**, Ung Heo, Sangha Jeon, and Jung Kim  
*The 39th Institute of Control, Robotics and Systems Annual Conference (ICROS 2024), 2024*
- [6] **Development of Compact Variable-Stiffness Actuator using Magnet Repulsion**  
Gunhee Park, Hyunkyu Park, **Woojong Kim**, and Jung Kim  
*The 39th Institute of Control, Robotics and Systems Annual Conference (ICROS 2024), 2024*
- [5] **Comparison of Normal Force Estimation Performance of the UV Marker-based Tactile Sensor According to Voronoi Area Data Utilization of Marker Images**  
**Woojong Kim**, Won Dong Kim and Jung Kim  
*The 38th Institute of Control, Robotics and Systems Annual Conference (ICROS 2023), 2023*
- [4] **Position Control of the Sensorized Flat Fabric Pneumatic Artificial Muscle (ffPAM)**  
**Woojong Kim**, and Jung Kim  
*The 16th Korea Robotics Society Annual Conference (KRoC 2021), Korea Robotics Society, 2021*
- [3] **Force Characterization and Dynamic Analysis of the Flat Fabric Pneumatic Artificial Muscle**  
**Woojong Kim**, and Jung Kim  
*2021 Korea Society of Biomechanics Annual Conference, 2021*
- [2] **Development of Backpack Loading Estimation Model under Different Locomotion Conditions (Level, Stair, Inclined) using Surface Electromyography (sEMG) and Ground Reaction Force**  
Seulki Kyeong, Junghoon Park, **Woojong Kim**, and Jung Kim  
*2017 Korea Institute of Military Science and Technology Annual Conference, 2017*
- [1] **Development of Backpack Loading Estimation Model using Surface Electromyography (sEMG) of Gastrocnemius During Walking and Running**  
Junghoon Park, **Woojong Kim**, Younghha Shin, and Jung Kim  
*Korea Society for Precision Engineering 2017 Spring Conference, 2017*

## TEACHING EXPERIENCE

---

### **Korea Advanced Institute of Science and Technology**

*Teaching Assistant*

Daejeon, South Korea

Mar. 2021 –Feb. 2025

- [ME488] Introduction to Biomedical Machine Technology, Spring 2021
- [ME203] Mechatronics System Design, Fall 2021, 2022, 2023
- [ME303] Mechanical Engineering Laboratory, Fall 2021 - Spring 2023
- [ME459] Introduction to Visual Intelligence, Spring 2022
- [ME400] Capstone Design, Spring 2024
- [ME106] Human and Machine, Fall 2024

## REFEREES

---

### **Jung Kim**

Department head and professor in Dept. of Mechanical Engineering, KAIST

(Email: jungkim@kaist.ac.kr)